Astronomical Bycatch: progress report

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1 Summary

The aim is to get light curves for the stars in archival images using the new extraction API in our private EXOTIC fork (`exotic.extract`). This report covers the first day of that work:

- Code to search and download raw data from the ESO Science Archive, and to find archive sequences that cover a known transit.
- A pipeline that plate-solves frames locally against Gaia DR3, runs `extract_sequence` on every usable star in the field, fits the target's transit and writes all light curves to JSON.
- **Validation:** on the HAT-P-32 b MicroObservatory night, the fitted mid-transit time agrees with EXOTIC's own validation result to within $\sim 1.5\sigma$.
- A wide-field ESO test night (POTS-1 b, ESO/MPG 2.2 m WFI) is downloaded, calibrated and plate-solved. Extraction is running.

2 Data sources

ESO Science Archive (raw frames). Metadata was queried through ESO's catalogue query service, and transiting planets through the NASA Exoplanet Archive. The two optical wide-field imagers, WFI and OmegaCAM, have 182 long single-pointing sequences between them (at least 40 frames over at least 2 hours). Only one catches a known transit: **POTS-1 b, WFI, night of 2007-05-10**. It has 310 Rc-band frames of 26 s over 8.2 h, covering the full transit with $\sim$2.5 h of data before and 2.8 h after. The night has 24 biases, 4 twilight flats and 10 screen flats; the night is 46 GB decompressed. POTS-1 is faint ($G = 17.3$) but its transit is deep, and the field is crowded with background stars.

MicroObservatory frames. David Lawley's repository of raw MicroObservatory frames covering seven hot-Jupiter targets. The HAT-P-32 night is byte-identical to EXOTIC's own sample data, which makes it a direct check against EXOTIC.

3 Code and storage

`access/` Searches the ESO archive, matches sequences to known transits, and downloads a target's frames and calibrations (`python -m access.fetch targets/<t>.json`). Downloads can be restarted. Each target is a small JSON file in `targets/`.

`pipeline/astrometry.py` Local plate solving against Gaia DR3, so no image is uploaded to an external service. Raw WFI and MicroObservatory frames carry no sky coordinates, and the MicroObservatory header pointing can be $\sim 0.3^\circ$ off.

`pipeline/microobs.py`, `pipeline/wfi.py` Pipelines for each instrument. WFI handles 8 detectors per frame: overscan trimming, master bias and flat, and a separate plate solve per detector.

`pipeline/store.py` JSON results, one file per sequence (and per detector). Each holds the observation details, per-frame times and airmass, and a light curve for every star keyed by Gaia DR3 ID. The layout maps onto the existing kdb+ schema (`sources`, `obs_sessions`, `observations`).

Data and results stay on the department compute server; nothing large is kept on laptops. Code is on the `huw-leggate/claude-md` branch of `creanero/bycatch`.

4 Results

4.1 HAT-P-32 b: validation

142 frames of 60 s. The plate solve matched 184 Gaia stars to 1.5 arcsec, and 83 stars were photometered.

Method. Each star's flux is measured in a circular *photometric aperture* centred on the star, with the sky level taken from a surrounding annulus. The aperture radius is set as a multiple of the stellar FWHM measured on the first frame. The target's relative light curve is its flux divided by the summed flux of the comparison stars. That curve is fitted with EXOTIC's least-squares `fit_transit`, after dividing out a linear trend with airmass. The fit has three free parameters:

- the mid-transit time T_{mid};
- the radius ratio R_p/R_s, the planet's physical radius divided by its host star's radius, which sets the transit depth (depth $\approx (R_p/R_s)^2$);
- a flux normalisation.

The orbit (period, scaled semi-major axis a/R_s, inclination) and the stellar limb darkening are held fixed. R_p/R_s has nothing to do with the photometric aperture radius. The aperture is only a measurement setting, and a well-behaved reduction should return the same R_p/R_s whatever aperture is used.

What is compared. We reduced the same night three times, with photometric apertures of 1.0, 1.4 and 1.9× the FWHM, and fitted each light curve separately (Table 1). Each fit is compared against two independent references:

1. **Timing.** EXOTIC's own full reduction of these same frames gives $T_{\mathrm{mid}} = 2458107.71380\,\mathrm{BJD_{TDB}}$. This is the value EXOTIC's validation tests check against. Matching it shows our extraction and timing reproduce EXOTIC's.
2. **Depth.** The literature value $R_p/R_s = 0.149$, from the HAT-P-32 b parameters in EXOTIC's configuration for this dataset (taken from the NASA Exoplanet Archive). EXOTIC's validation tests don't check R_p/R_s: in its fit R_p/R_s is degenerate with the orbital inclination, so it isn't stable enough to test against. So depth is compared with the published value instead.

Outcome.

- **Timing:** all three fitted mid-transit times agree with EXOTIC's to within $\sim 1.5\sigma$ (offsets of -0.6 to $+4.0$ min, with uncertainties of about 2.5 min). The spread between apertures is comparable to the uncertainties.
- **Depth:** the fitted planet-to-star radius ratio is 0.151 to 0.157, 1–5% above the literature 0.149. The most likely cause is the limb darkening, held at generic approximate coefficients rather than ones computed for HAT-P-32 in this filter. That changes the transit's shape, and with it the depth the fit needs. It has little effect on T_{mid}, which depends on the transit's symmetry.

Table 1: HAT-P-32 b transit fits for three photometric aperture sizes. Each row is a separate reduction of the same 142 frames. *Aperture radius* is the photometric aperture in pixels and in multiples of the stellar FWHM. *Scatter* is the point-to-point scatter of the relative light curve. T_{mid} *offset* is the fitted mid-transit time minus EXOTIC's result for this night (2458107.71380 $\mathrm{BJD_{TDB}}$). *Fitted* R_p/R_s is the planet-to-star radius ratio from the fit; the literature value is 0.149.

Photometric aperture radius (×FWHM)	(px)	Scatter	T_{mid} offset (min)	Fitted R_p/R_s (planet / star)
1.0	2.5	0.53%	-0.6 ± 2.6	0.151 ± 0.003
1.4	3.5	0.48%	$+2.1 \pm 2.4$	0.157 ± 0.003
1.9	4.7	0.57%	$+4.0 \pm 2.8$	0.156 ± 0.004

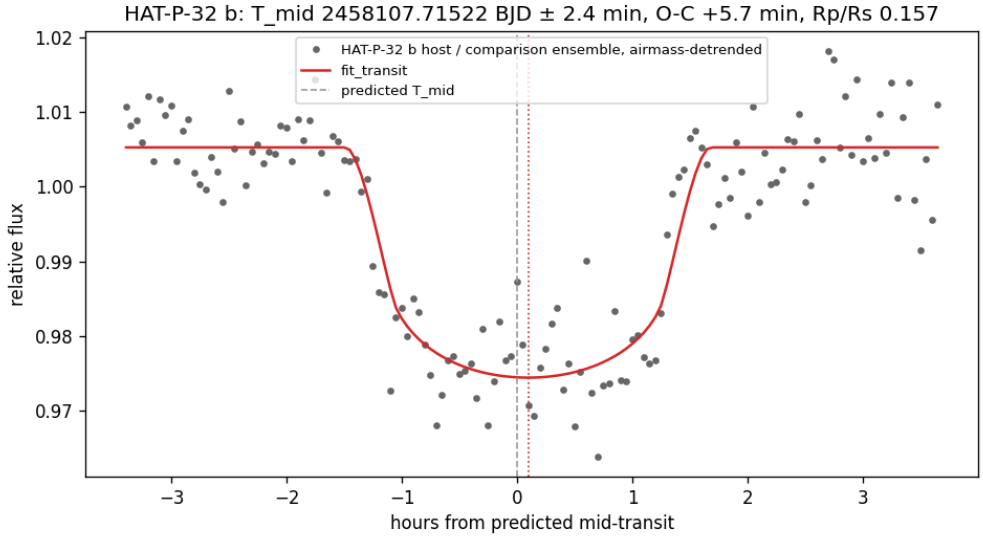


Figure 1: HAT-P-32 b relative light curve (grey points) with the fitted transit model (red), from the reduction with a photometric aperture radius of 1.4×FWHM (3.5 px). The x-axis is hours from the mid-transit time predicted by the published ephemeris. The fitted T_{mid} is 5.7 min after that prediction and 2.1 min after EXOTIC's result for this night; the fitted R_p/R_s is 0.157 (Table 1).

4.2 WASP-36 b

55 frames of 60 s. Plate solving and frame alignment worked, but the target ($G = 12.6$) is limited by sky noise: the best scatter is 2.3% per point, with a photometric aperture radius of 1.0×FWHM. The transit is ∼1.8% deep, and a 66-minute gap leaves only 3 frames before ingress. The fit (Figure 2) puts T_{mid} $+26 \pm 43$ min from the ephemeris prediction and a planet-to-star radius ratio $R_p/R_s = 0.09 \pm 0.06$ (literature 0.134). Both are consistent with expectations, but too uncertain to be useful. For comparison, the first reduction used the default photometric aperture radius of 1.9×FWHM (4.3 px). It gave 6.2% scatter and T_{mid} -7 ± 65 min, $R_p/R_s = 0.16 \pm 0.11$. Scatter fell steadily with smaller apertures, as expected for a faint star dominated by sky noise.

4.3 POTS-1 b, WFI (in progress)

- **Calibration:** gain and read noise were measured for each detector from bias and flat pairs, because the headers record gain for only two detectors and read noise for none. Gain came out at 1.94–2.28 $\mathrm{e^-/ADU}$ and read noise at 5.6–6.2 $\mathrm{e^-}$, consistent with WFI's documented values.
- **Astrometry:** all 8 detectors plate-solved, with 36–163 Gaia stars each and residuals of 0.06–0.11 arcsec. POTS-1 is on chip 3.

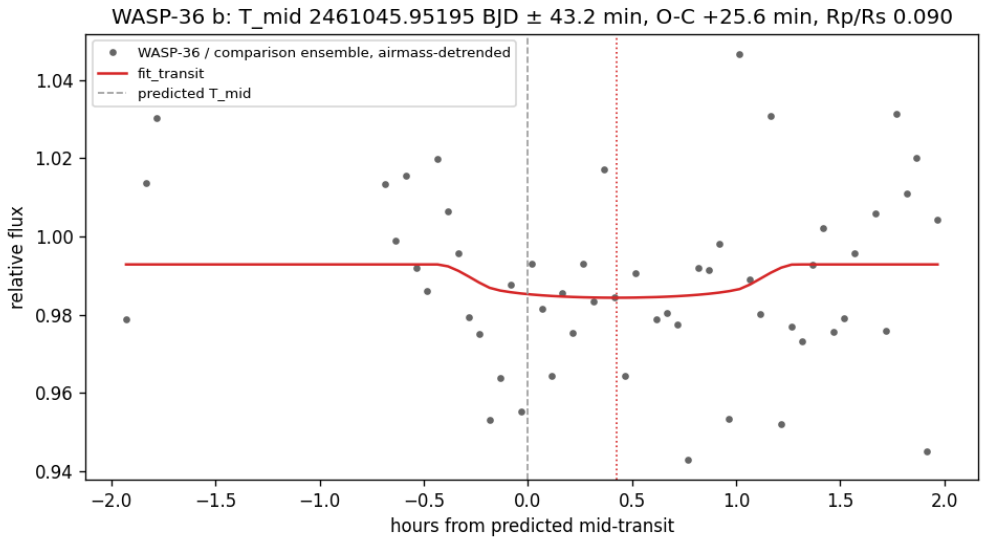


Figure 2: WASP-36 b relative light curve (grey points) with the fitted transit model (red), from the reduction with a photometric aperture radius of 1.0×FWHM (2.3 px). The dashed grey line is the predicted mid-transit time and the dotted red line the fitted one. The scatter is comparable to the transit depth.

First extraction (failed). 801 stars down to $G = 18.5$ were photometered across all 310 frames, with a photometric aperture radius of 1.9×FWHM (10.1 px). The result is shown for information in Figure 3. The target's relative light curve scatters by 27% from point to point. That's far above the ∼1% photon noise expected for a $G = 17.2$ star in a 26 s exposure on a 2.2 m telescope, and it swamps the ∼2.5% transit. Every star on the chip is affected: scatter is 1.8% at $G = 12$, 4.4% at $G = 15$ and over 20% at $G \geq 17$, against predicted uncertainties of 0.06–0.9%. Star tracking was fine: the target stays within 0.4 px of the comparison stars.

There were two causes:

- **Sky background (EXOTIC).** `skybg_phot` takes the sky level as the statistical *mode* of the pixel values in a box around each star. The box includes the star itself, not just the annulus. On raw integer frames (MicroObservatory) the mode works, which is why HAT-P-32 validates. On bias- and flat-corrected frames, which are floating point, almost every value is unique, so the "mode" is an arbitrary pixel. For the target it jumped between 28 and 62 ADU from frame to frame, against a true sky of ∼50 ADU. Over the 320-pixel aperture that is a 20–30% flux error. Workaround: round calibrated frames to whole ADU before extraction. That adds 0.3 ADU of rounding noise against 7 ADU of sky noise.
- **Fixed orbit (ours).** The NASA Exoplanet Archive has no tabulated a/R_s for POTS-1 b, and the code defaulted it to zero, so the transit model was degenerate (hence the infinite uncertainty in Figure 3). It is now derived from the semi-major axis and the stellar radius.

Corrected extraction. Re-running with both fixes, at photometric aperture radii of 1.2 and 1.9×FWHM. Results to follow.

5 Issues and next steps

- **Speed:** centroid fitting in `extract_frame` takes ∼70 ms per star per frame, which dominates run time (about 55 s per WFI frame for 800 stars). Running in parallel works around it for now. This is a useful benchmark for the GPU work in EXOTIC.
- **EXOTIC sky background:** `skybg_phot`'s modal estimate fails on calibrated, floating-point frames (Section 4.3). We work around it by rounding, but the proper fix belongs in EXOTIC:

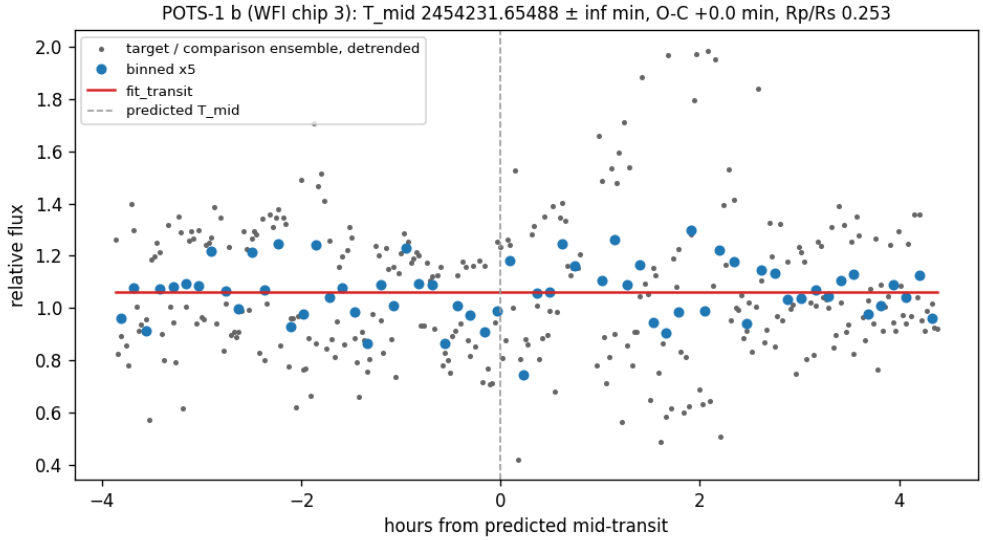


Figure 3: First, failed extraction of POTS-1 b (WFI chip 3), shown for information. Grey points are the target's relative flux per frame, blue points are 5-frame bins, and the red line is the failed fit. The 27% point-to-point scatter comes from the unstable sky-background estimate on calibrated (floating-point) frames. The fit is meaningless because a/R_s was zero. Both problems are fixed in the re-run.

estimate the sky from the annulus pixels only, with a sigma-clipped median, so it works whatever the data type. EXOTIC's own pipeline is affected whenever it applies darks or flats.

- **Photometric aperture size:** the default FWHM-scaled aperture is too large for faint, sky-limited stars. It should be tuned per star, or chosen by sweeping, for bycatch at scale.
- **Limb darkening:** fixed approximate coefficients bias the fitted R_p/R_s. Timing is unaffected.
- **Next:** finish POTS-1; run the remaining MicroObservatory targets; look for variability among the field stars in the JSON results; then scale the search to OmegaCAM and the other WFI sequences.